

Quiz 1 Results for Rakshanda Hassan Abhinandan

❗ Correct answers are hidden.

Score for this attempt: 8 out of 10

Submitted Jan 18 at 2:45pm

This attempt took 326 minutes.



Question 1

1 / 1 pts

According to the university's academic integrity policies and 11785 course policies, which of the following practices are **NOT** allowed in this course? Select all that apply.

- ☒ Discuss quiz solutions with another student before the deadline
- ☒ Creating a piazza post asking for help debugging code without prior attempts to debugging it yourself.
- ☐ Asking a TA for help debugging your code
- ☐ Discussing concepts from class with another student
- ☒ Posting code in a public post on piazza
- ☐ Helping another student debug their code

See <https://www.cmu.edu/policies/student-and-student-life/academic-integrity.html> (<https://www.cmu.edu/policies/student-and-student-life/academic-integrity.html>). Other than these restrictions, we highly encourage you to discuss course concepts with other students! Post (reasonable) questions on Piazza, and answer others. Learning is most fun with friends :)



Question 2

1 / 1 pts

Is the following statement true or false? Hebbian learning allows reduction in weights and learning is bounded.

Slide: lec 1, "Hebbian Learning" slide 66

- ☐ True
- ☒ False

If neuron x repeatedly triggers neuron y, the synaptic knob (Weight) connecting x to y gets larger. Hence the weight only increases and a mechanism for weight reduction is not given. Also, the upper bound to

which the weight increases to is not defined in the learning, making it unbounded.



Question 3

1 / 1 pts

Which of the following are **NOT** part of Aristotle's laws of association? (select all that apply)

Hint: See lec 1: Slides on "Associationism" 32-37

- ☐ Thoughts about one event or thing triggers thoughts about related events or things.
- ☐ Thoughts about one event or thing triggers thoughts about events or things that have opposite qualities
- ☒ Mental willpower is used to associate unrelated events or things
- ☒ Associations are formed from logical deduction.
- ☐ Events or things near the same place and time are associated.

In Lecture 1, Aristotle's laws of association are presented as basic mechanisms by which ideas become linked through experience. These laws focus on relationships such as proximity in time or space, similarity, contrast, and repeated co-occurrence. Importantly, they describe associations as arising naturally from how cognition works, rather than from deliberate reasoning or conscious effort. Options that appeal to mental willpower or logical deduction may sound plausible in everyday terms, but they fall outside the four associationist laws explicitly listed in the lecture.

Incorrect



Question 4

0 / 1 pts

How does the number of weights in an XOR network with arbitrarily many hidden layers and threshold logic gates grow with the number of inputs to the network?

See lec 2: Slides on "Optimal depth" and "Network size" 113-123

- ☐ Polynomial but faster than linear
- ☐ Linear
- ☒ Between polynomial and exponential
- ☐ Exponential or faster

In the XOR circuit shown in the slides, the number of perceptrons grows linearly with input, and each perceptron has only two inputs and, hence, a fixed number of weights (2). Thus, the number of weights too grows linearly with input



Question 5

1 / 1 pts

What is the implication of Shannon's theorem on network size (as a function of input size) for Boolean functions?

Slide: lec 2, slide 71-72

- ☐ Only a relatively small proportion of functions require exponentially sized inputs
- ☐ Most functions require polynomial sized networks.
- ☐ All functions require exponential-sized networks
- ☒ Nearly all functions require exponential-sized networks

Shannon's theorem uses a counting argument: the number of possible Boolean functions on n inputs grows as 2^{2^n} , while the number of networks of polynomial size grows much more slowly. As a result, although some Boolean functions admit compact representations, *most* Boolean functions cannot be represented without an exponential number of units. This is why the lecture emphasizes that nearly all Boolean functions require exponential-sized networks.

For those interested in a deep dive, here is Shannon's paper, around page 76 (access using CMU account):

<https://ieeexplore.ieee.org/document/6771698> (<https://ieeexplore.ieee.org/document/6771698>)

⋮

Question 6

1 / 1 pts

What is the fewest number of neurons needed (including any output layer neurons) for a network to implement the truth table shown by the following Karnaugh map? (numeric answer, int and float are both fine)

WX \ YZ				
	00	01	11	10
00				
01				
11				
10				

Hint: lec 2, "Reducing a Boolean Function". Slide 47 - 50

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The Boolean function is $O = WX + YZ + XZ$. The reduced DNF has 3 terms, so the Boolean network uses 3 hidden units plus 1 output unit (total 4)



Question 7

1 / 1 pts

Suppose the data used in a classification task is 10-dimensional and positive in all dimensions. You have two neural networks. The first uses threshold activation functions for hidden layers, and the second uses softplus activation functions for hidden layers. In both networks, there are 2000 neurons in the first hidden layer, 8 neurons in the second hidden layer, and a huge number of neurons for all later layers. It turns out that the first network can never achieve perfect classification. What about the second network?

Hint: A layer with 8 neurons effectively projects the data onto a 8-dimensional surface of the space. The input is 10-dimensional.

Lec 2 slides 138-150



Assuming that layers 3 and above are so expressive that they never bottleneck the flow of information, the second network will be able to achieve perfect classification.



It might fail for some data sets, not only because a bottleneck can occur, but also because the 2000 neurons in the first hidden layer could bottleneck the flow of information if the classification task is sufficiently complex.



It too is guaranteed to be unable to achieve perfect classification.



It might fail for some data sets, since the 8 neurons in the second hidden layer could bottleneck the flow of information. In that case, the sizes of layers 3 and above don't matter.

Answer: B. The 8-neuron layer reduces the dimensionality of the data from 10 to 8. This results in a loss of dimensionality, and potentially, of information.

Regardless of the activations, such a loss of dimensionality can result in loss of key information which prevent perfect classification, unless the data themselves actually live in an 8-dimensional surface in the 10-dimensional space. The fact that the threshold activation network fails to achieve perfect accuracy in spite of having a large number of initial feature detectors suggests that the data do not in fact live on an 8-dimensional surface, although we cannot be certain of it.

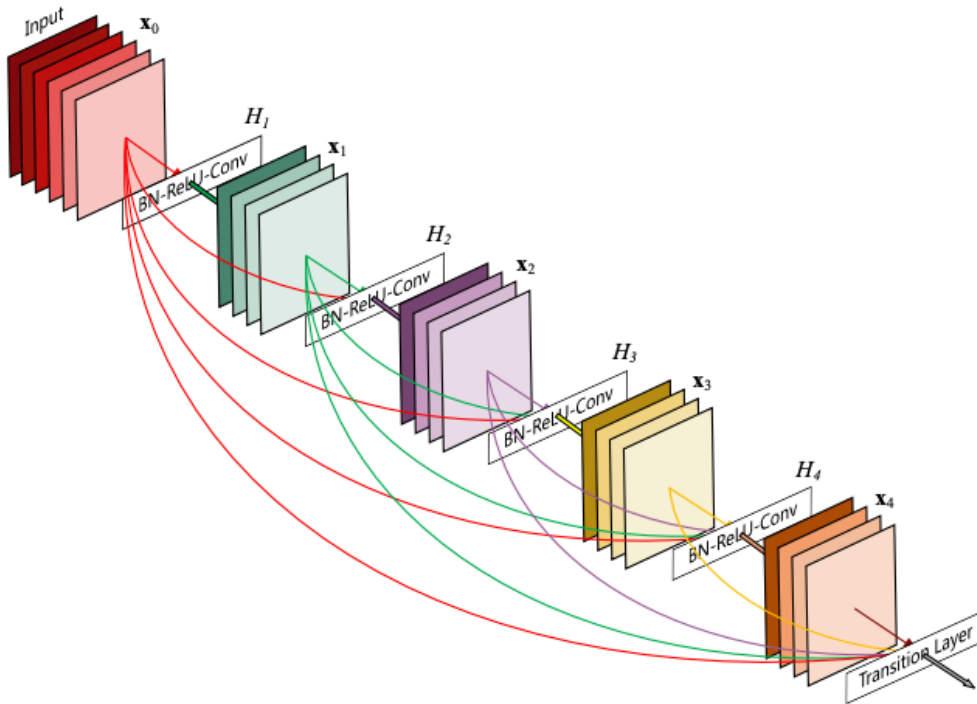
Incorrect



Question 8

0 / 1 pts

Densenet (2017) is a CNN network architecture that achieved SoTA results with fewer params than many of its contemporaries. Its main idea was the 'dense block': a block where each layer output is concatenated into the input of the downstream layers.



Above is a diagram of a dense-block. Each " H_i " consists of Batchnorm->ReLU->Convolution. Each H_i outputs a " x_i " (one stack of squares in the image) .

What is the depth of the dense block above? (numeric answer, int and float are both fine)

Assume that "BN+RELU+CONV" is a single layer. The square planes simply represent the data flowing down between layers and are not layers themselves.

(Note: for the definition of network depth, see the lecture 2 recording)

Slide: lec 2, "Deep Structures"

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Depth = length of the longest path (counting layers along that path). Here the longest chain is $\text{Input} \rightarrow H_1 \rightarrow H_2 \rightarrow H_3 \rightarrow H_4 \rightarrow \text{Transition}$, which includes 4 BN+ReLU+Conv layers (H_1 - H_4) plus the Transition layer, for a total depth of 5.



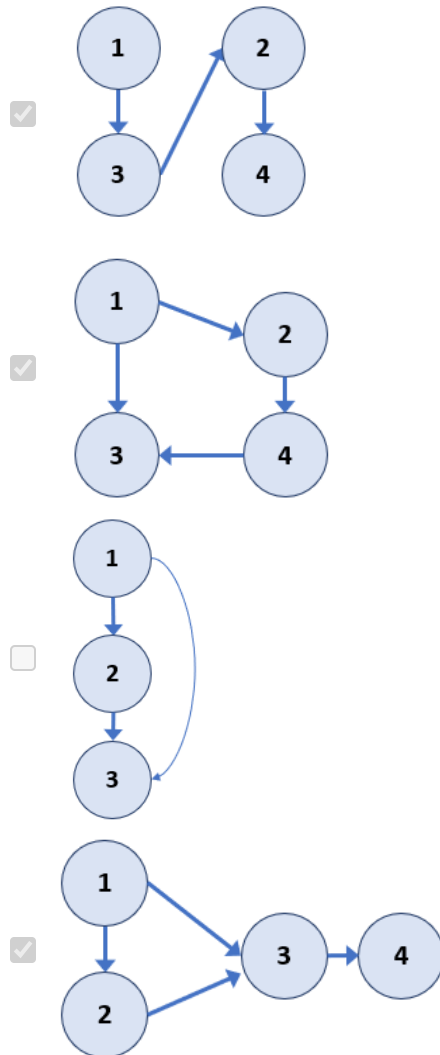
Question 9

1 / 1 pts

Which of the following NNs have a depth of 3? (Choose all that apply)

(Note: for the definition of network depth, see the lecture 2 recording)

Hint: lec 2, "Deep Structures", *Slides*: 17-18



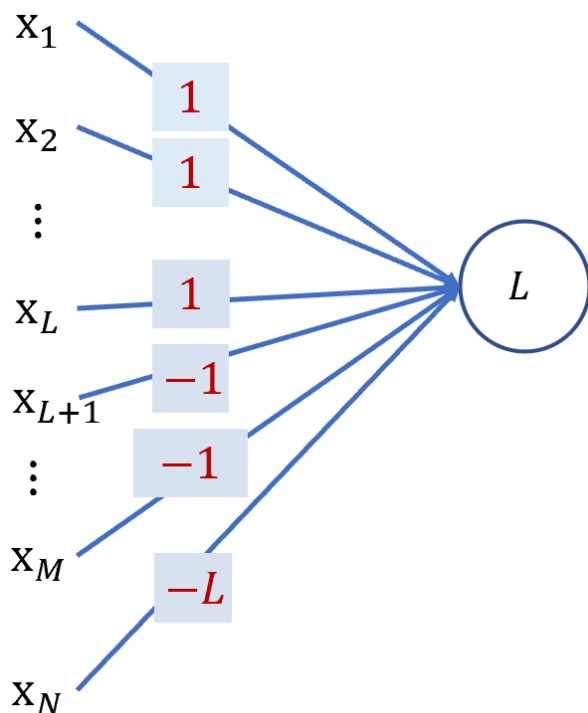
Depth is defined as longest path from source to sink.

- a) longest path: 1 → 2 → 3 (Depth 2)
- b) longest path: 1 → 2 → 4 → 3 (Depth 3)
- c) longest path 1 → 3 → 2 → 4 (Depth 3)
- d) longest path: 1 → 2 → 3 → 4 (Depth 3)



Question 10

1 / 1 pts



What boolean function does the above perceptron represent?

(Hint: notice that $M = N - 1$)

Hint: Lecture 2 Slides: "Perceptron as a Boolean gate", Slides 26-30

- ☐ $\left(\bigvee_{i=1}^L X_i \right) \wedge \left(\bigwedge_{j=L+1}^M \bar{X}_j \right) \vee (X_N)$
- ☐ $\left(\bigwedge_{i=1}^L X_i \right) \wedge \left(\bigwedge_{j=L+1}^M \bar{X}_j \right) \vee (X_N)$
- ☐ $\left(\bigwedge_{i=1}^L X_i \right) \vee (X_M) \wedge (\bar{X}_N)$
- ☒ $\left(\bigwedge_{i=1}^L X_i \right) \wedge \left(\bigwedge_{j=L+1}^N \bar{X}_j \right)$

There are N weights in total. Also notice that $M = N - 1$. To cross the threshold, we need a total contribution of L . This means that inputs X_1 to X_L need to be contributed, also not a single negative input (X_{L+1} to N) can be contributed.

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